CONSTITUTIONAL PATENT PROTOCOL (ARTICLE II, SECTION 1, CLAUSE 8)

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PRO SE PATENT APPLICATION FOR LETTERS PATENT

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SYSTEM AND METHOD FOR SELF-HEALING OPTICAL COORDINATE ANCHORING AND VERTICAL OFFSET CALIBRATION IN HIGH-RADIX PHYSICAL REGISTRIES

PROVISIONAL PATENT DISCLOSURE

FIELD OF THE INVENTION

This invention relates generally to optical character recognition, digital signal processing, document imaging, and data reconstruction. More specifically, this invention relates to an optical decoding system and auto-anchoring algorithm that dynamically locates and self-calibrates the coordinate origin of a printed high-radix dot matrix on paper, compensating automatically for physical feed slippage, scanner dead zones, print driver margins, and image rotation.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, world-wide, non-exclusive

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BACKGROUND AND TECHNICAL CHALLENGE

High-density visual cryptographies and paper-digital registers pack thousands of data-carrying dots into standard paper mediums. However, when these pages are printed, standard operating system print drivers and spoolers frequently apply custom, arbitrary margins or vertically center truncated images on the 11-inch sheet. Furthermore, when the physical page is scanned back via office flatbeds, document feeders, or mobile phone cameras, the resulting digital file suffers from rotation, vertical/horizontal shifts, shadow clipping, and feed slippage.

If a decoder relies on absolute page coordinates (e.g., assuming the first row of dots starts strictly at a static vertical margin), any slight shift or paper misfeed will completely misalign the sampling grid. This results in sampling adjacent pixels or background white space, causing immediate data-corruption or decode failure.

A robust, self-healing visual alignment processor is needed that can automatically, dynamically scan the ingested image, identify the vertical and horizontal start boundaries of the dot grid, and anchor the sampling coordinates relative to that dynamic origin.

THE AUTO-ANCHORING & CALIBRATION MATHEMATICS

The present invention solves these problems by providing an Auto-Anchoring Visual Grid Calibration Processor.

The decoder ingests a scanned RGB image of height H and width W. Instead of starting at a hardcoded coordinate, the processor executes a two-phase dynamic anchor scan:

Phase 1: Row-by-Row Vertical Origin Scanning (Y_{origin})

To locate the absolute vertical start of the grid, the processor scans down the page row-by-row, sampling the pixels horizontally every 10 pixels.

To prevent false-positives triggered by flatbed glass dirt, paper fibers, or scanner border shadows, the processor implements a **consecutive dual-pixel validation threshold**:

- For each row index $y \in [0, H-2]$, the system checks if there exists a horizontal coordinate x where both pixels (x, y) and $(x, y+1)$ are non-background (RGB values < 240 in all channels).
- The first row index y that passes this dual-pixel validation is locked as the absolute vertical origin Y_{origin}:

$$Y_{\text{origin}} = y_{\text{detected}}$$

Phase 2: Horizontal Left-Edge Detection (X_{origin})

Once the vertical origin is locked, the processor scans left-to-right along the single row Y_{origin} to identify the first non-background pixel. This coordinate is locked as the absolute horizontal origin X_{origin}:

$$X_{\text{origin}} = x_{\text{detected}}$$

Phase 3: Dynamic Grid Alignment

All sampling coordinate centers $(X_{\text{sample}}, Y_{\text{sample}})$ for the entire multi-thousand dot grid are then calculated relative to the dynamically anchored coordinate origin $(X_{\text{origin}}, Y_{\text{origin}})$ and the measured **dot_size** parameter:

$$X_{\text{sample}}(dx) = X_{\text{origin}} + (dx \times \text{dot_size}) + \lfloor \text{dot_size}/2 \rfloor$$

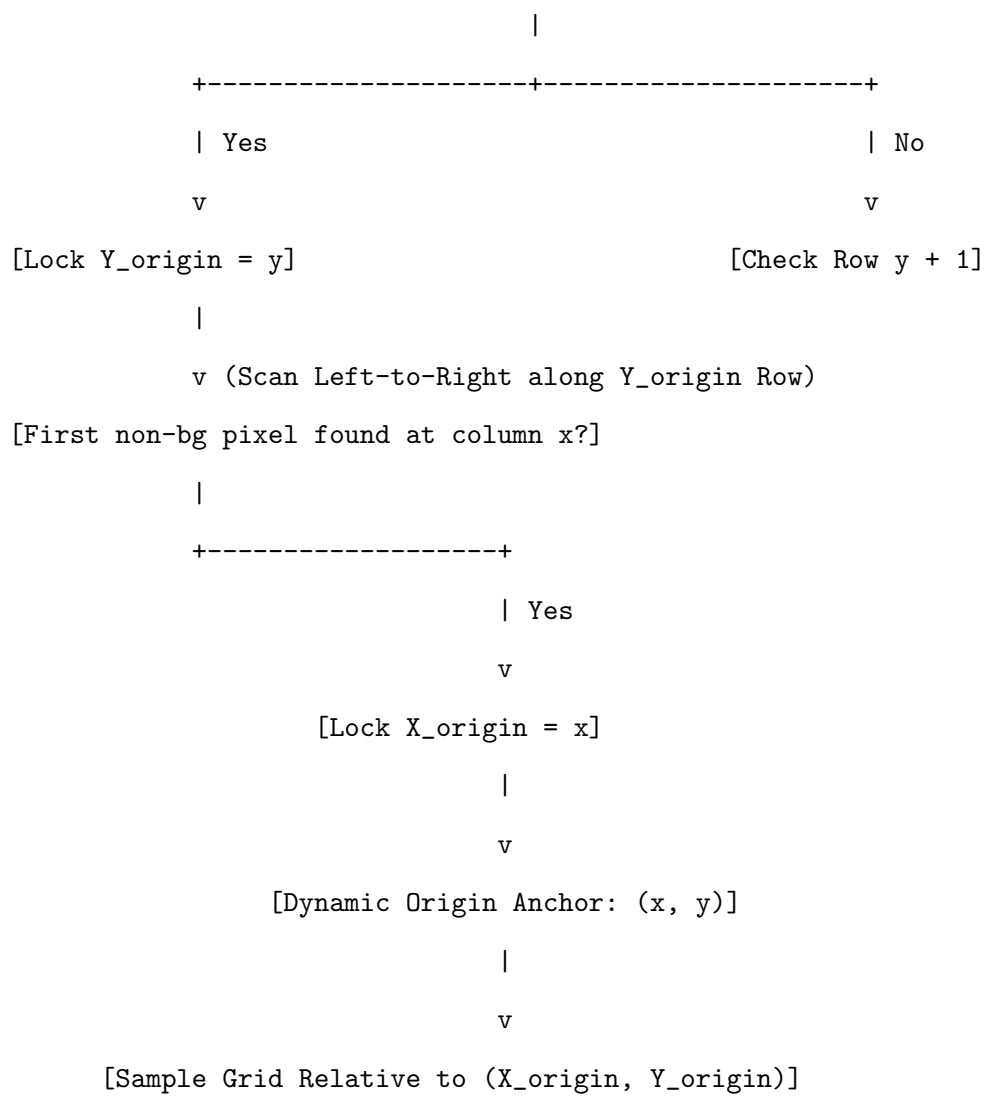
$$Y_{\text{sample}}(dy) = Y_{\text{origin}} + (dy \times \text{dot_size}) + \lfloor \text{dot_size}/2 \rfloor$$

This completely decouples the decoder from any absolute, machine-specific coordinates, page sizes, or margins of the scanned page.

THE CALIBRATION PROCESS SCHEMATIC

The diagram below maps out the auto-anchoring origin detection flow:

```
[Ingested Scanned Image (W x H)]
               |
               v (Scan Row-by-Row Downward)
[Check Row y] -> Sample every 10 px -> [Consecutive non-bg pixels found?]
```



PROVISIONAL PATENT CLAIMS

1. **An optical decoding processor for automatically calibrating and reconstituting a physical high-radix dot matrix from a scanned document container, comprising:**

   - an optical input interface configured to ingest a scanned page and output a 24-bit RGB pixel matrix of height H and width W;
   - a processor electrically coupled to the optical input interface and configured to execute an auto-anchoring origin calibration by:
     - scanning down the 24-bit RGB pixel matrix row-by-row horizontally to identify a first row Y_{origin} containing at least one consecutive pair of non-background pixels,

skipping scanner border noise and dust;

- scanning left-to-right along the identified Y_{origin} row to locate a first non-background pixel at column X_{origin}, thereby establishing a dynamic visual coordinate origin $(X_{\text{origin}}, Y_{\text{origin}})$ that automatically compensates for vertical print shifts, paper slippage, and scanner clipping;
- scanning horizontally from X_{origin} to measure a consecutive non-background pixel width, and mapping the measured pixel width to a discrete dot-pixel size robustness metric; and
- sampling the pixels of the RGB matrix at coordinate coordinates calculated relative to the dynamic visual coordinate origin $(X_{\text{origin}}, Y_{\text{origin}})$ and the measured dot-pixel size to extract convolved 12-bit color-state data; and

- wherein the optical decoding processor is entirely decoupled from absolute physical coordinates, page sizes, or margins of the scanned document container.

Respectfully submitted,

/Reinaldo Javier Aguiar Marcano/
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